La quatrième mélodie for violin and percussion

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Abstract

This article discusses the compositional processes behind *La quatrième mélodie* (2024–25) for violin and percussion. The title refers to a proposed "fourth melodic situation" in Joël-François Durand's recent music, which features interference patterns caused by frequencies in close proximity. The precompositional material of the piece samples a mathematical representation of this phenomenon. Despite this, the music itself never fully reproduces the fourth melodic situation; it approaches it only asymptotically.

1 Introduction

1.1 The "fourth melody"

The title *La quatrième mélodie* relates to Joël-François Durand's article "Melody—Three Situations: *Un Feu distinct*, *La Terre et le feu*, *Athanor*" (Durand, 2006b). In this article, the composer describes three situations in which melodies serve two main functions, creating an immediate affect and carrying a form-building potential. In a self-interview included in the same volume, Durand distinguishes "melody" from "line" by its presence: whereas a line is "a general movement" perceived "in a kind of statistical way," a melody "is imbued with a specific affect; the stronger its presence, the more memorable it is" (Durand, 2006a, pp. 71–72). This presence also underlies its form-building role, as he sees the melody as "a structure-bearing unit that has enough of a singular character to acquire strong presence in its individuality" (Durand, 2006a, p. 72).

In the three works, this long-term formal function manifests itself in different manners. The melodic lines are interpreted as continuous contours in pitch–time space, which are then read at the attack points of new rhythmic patterns in *Un Feu distinct* (Durand, 2006b, pp. 100–105); the original melody is cut into syntactic units that are reassembled in constantly varying ways in *La Terre et le feu* (Durand, 2006b, pp. 105–112); and the melody is projected across large sections of the piece by altering its temporal proportions in *Athanor* (Durand, 2006b, pp. 112–121).

Durand's recent work appears to introduce a fourth melodic situation, whose hallmark is the perception of interference patterns arising from frequencies in close proximity. Durand notes that perceiving the beats requires the listener to shift attention away from the tones that produce them, moving between the two layers as if listening to a new type of polyphony (Durand, 2026). This oscillation of attention underpins the form-building potential of the fourth melodic situation, carrying it beyond its initial appearance as a local phenomenon.

The fourth melodic situation's short- and long-term realizations in Durand's recent work, and its relation to the other three, deserve separate scholarly attention.

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Figure 1: An example of the fourth melody

2 Compositional process

La quatrième mélodie is scored for violin and percussion. The percussion includes a vibraphone and two steel spring coils, which are placed on a resonant surface such as the head of a bass drum. The piece lasts around four minutes.

2.1 The asymptotic narrative

The fourth melodic situation, hereafter referred to as the "fourth melody," does not explicitly appear in the piece. The relation between the fourth melody and the musical choices in the piece is asymptotic: the music always tends toward it without reaching it. This relationship takes various forms, from the obfuscation of what would otherwise be a clear fourth melody to a particular asymptotic relationship that appears in the refrains, discussed below (Figure 3).

To see how obfuscation works, it helps to compare an instance of a fourth melody with its asymptotic reflection in *La quatrième mélodie*. Figure 1, an excerpt from Durand's *Geister, schwebende Geister…* (Durand, 2020), provides a concrete example of the fourth melody. The interference patterns have several sources. The frequency differences between the solo viola's third and fourth strings, and among the English horn, vibraphone, and ensemble viola, produce beating patterns of 3 Hz and 7 Hz. These are then reinforced by the septuplet and triplet figures. The immediate affect of the fourth melody is further secured by the *non vibrato* "policy" across the instruments and by the absence of any local timbral or pitch alterations.

The vibraphone's steady A_4 in *Geister, schwebende Geister…* is the reference pitch against which the other instruments' closely tuned frequencies beat. In *La quatrième mélodie*, the instrument's mechanical vibrato assumes an opposite role and obscures the interference patterns rather than anchoring them (Figure 2). The percussionist is asked to actively alter the vibrato rate, while the violinist uses a slow vibrato pattern that also serves as a nonlinear *glissando*. Other pitch alterations and timbral properties, such as the combination of *sul ponticello* and bow pressure (notated via darker gradients under the noteheads, see Appendix A for details), and slightly lighter touch pressure (denoted via gray noteheads) on the violin, contribute to this obfuscation. As a result, a potential interference pattern between the violin's G-quarter-sharp$_3$ and the vibraphone's G_3 is hindered from fully emerging (Figure 2).

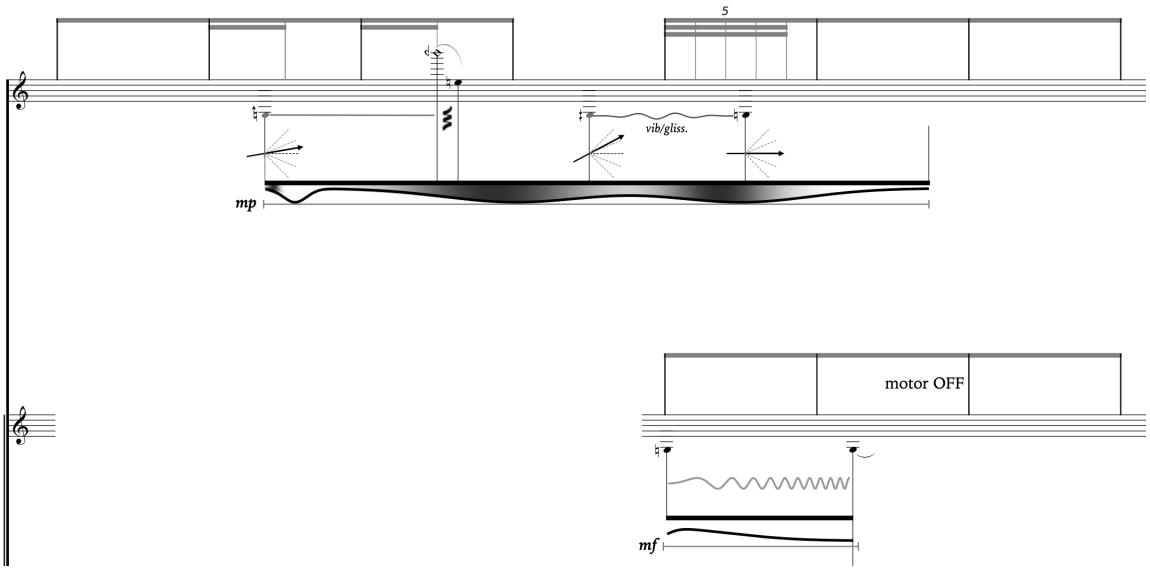


Figure 2: Obfuscation of the fourth melody

The flow of *La quatrième mélodie* is delineated by a refrain. The Durand excerpt in Figure 1 is the source for the refrain exemplified in Figure 3. The refrain provides a different aspect of the asymptotic relationship with its source. Here, we clearly hear the triplet and septuplet rhythms, and the source's pitch specificity is reflected in a particular vibrato depth. However, this appears to be a mere rhythmic emulation of the interference patterns: even if the violinist aimed to play the exact frequencies (they are not expected to do so), the resultant beating pattern would be twice as fast as the written rhythm, since the tempo is ♪ = 30 rather than ♪ = 60. The recurring refrains do shape the overall form, but this has no relation to the psychoacoustic structuring that is in effect in the original fourth melody: the emergence of a musical narrative that is supported by the listener's ever-moving attention between the sounding pitches and their beating patterns.

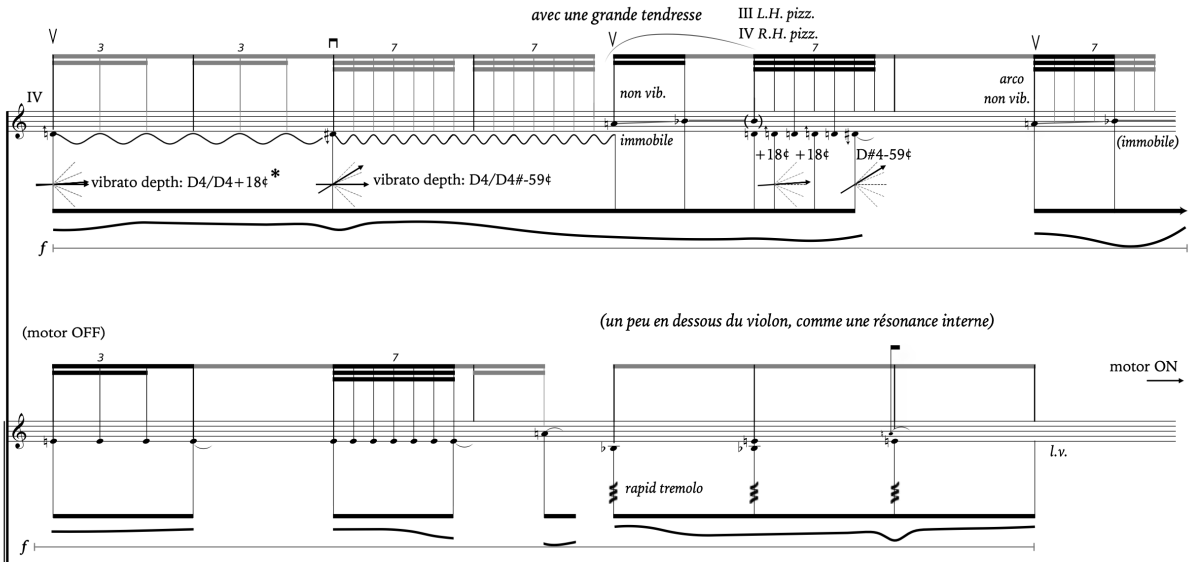


Figure 3: The refrain

2.2 Precompositional process

The goal of the precompositional process is to sample a number of arbitrary interference patterns and use each sample as the standard deviation of a normal distribution, from which a random value is drawn and mapped onto a notational parameter (see Figure 5 for an example).

2.2.1 Generating the interferences

Interference patterns are created by additive synthesis between a baseline sinusoidal signal $f(t) = \sin(\omega_0 t)$, with $\omega_0 = 1.0$, and three neighboring signals:

$$g(t) = \sin(\omega_1 t), \quad h(t) = \sin(\omega_2 t), \quad k(t) = \sin(\omega_3 t),$$

with $\omega_1 = 1.149$, $\omega_2 = 1.867$, and $\omega_3 = 0.968$. Summing each with the baseline yields three interference patterns, $f(t) + g(t)$, $f(t) + h(t)$, and $f(t) + k(t)$ (Figure 4). All signals have unit amplitude and zero phase throughout the process.

2.2.2 Assignment of the signals to notational parameters

In the implementation, the interference patterns are computed as complex signals:

$$z(t) = e^{i\omega_0 t} + e^{i\omega_1 t}, \quad w(t) = e^{i\omega_0 t} + e^{i\omega_2 t}, \quad v(t) = e^{i\omega_0 t} + e^{i\omega_3 t}.$$

By Euler's formula, $e^{i\theta} = \cos(\theta) + i\sin(\theta)$, so the imaginary part of each signal is one of the interference patterns above, e.g. $\operatorname{Im} z(t) = f(t) + g(t)$, and its real part is the corresponding sum of cosines. Writing $\tilde{f}(t) = \cos(\omega_0 t)$, and likewise $\tilde{g}$, $\tilde{h}$, and $\tilde{k}$,

$$\operatorname{Re} z(t) = \tilde{f}(t) + \tilde{g}(t), \quad \operatorname{Re} w(t) = \tilde{f}(t) + \tilde{h}(t), \quad \operatorname{Re} v(t) = \tilde{f}(t) + \tilde{k}(t).$$

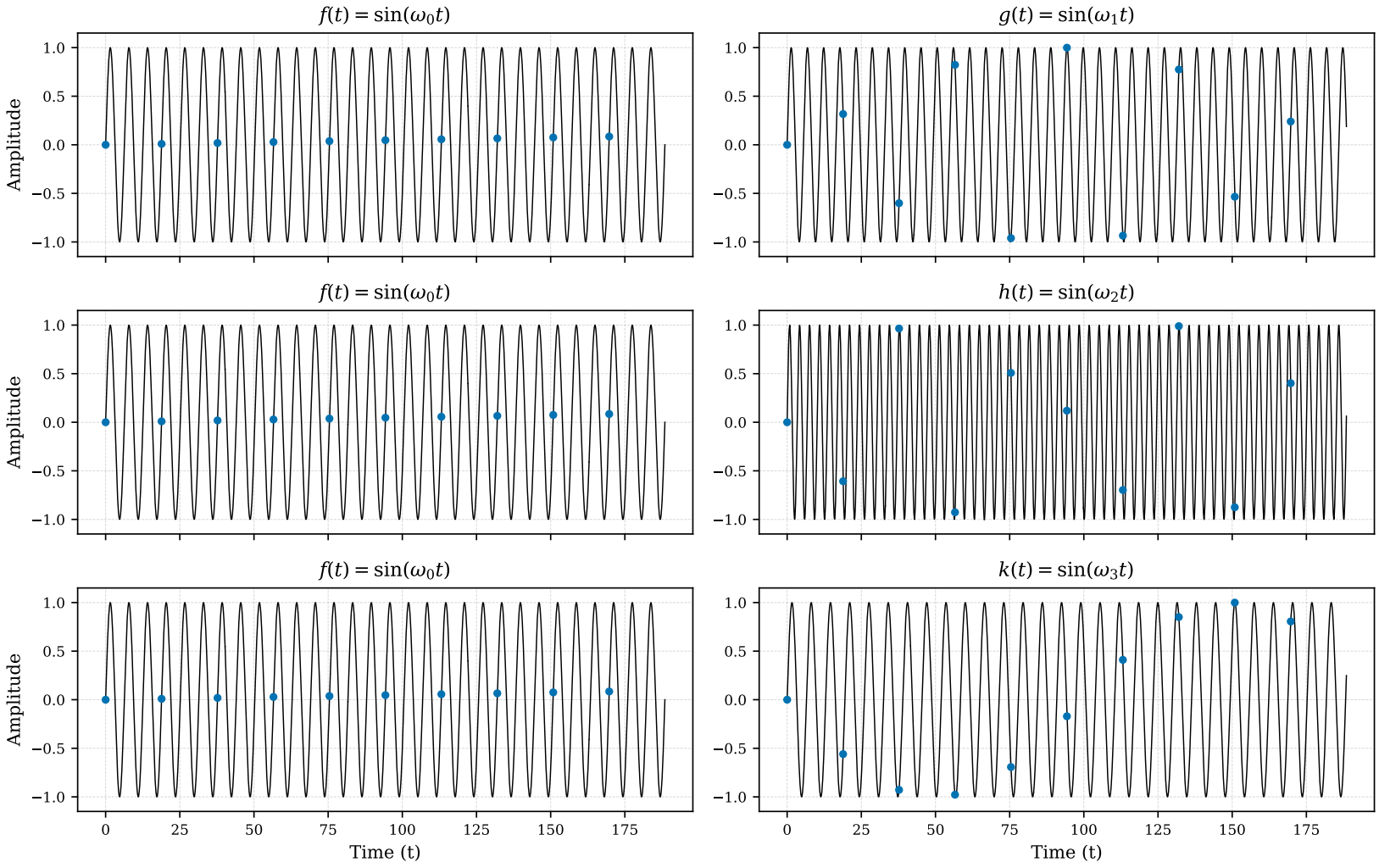


Figure 4: Generation of the interference patterns: (a) the baseline $f(t)$ and the three neighboring signals

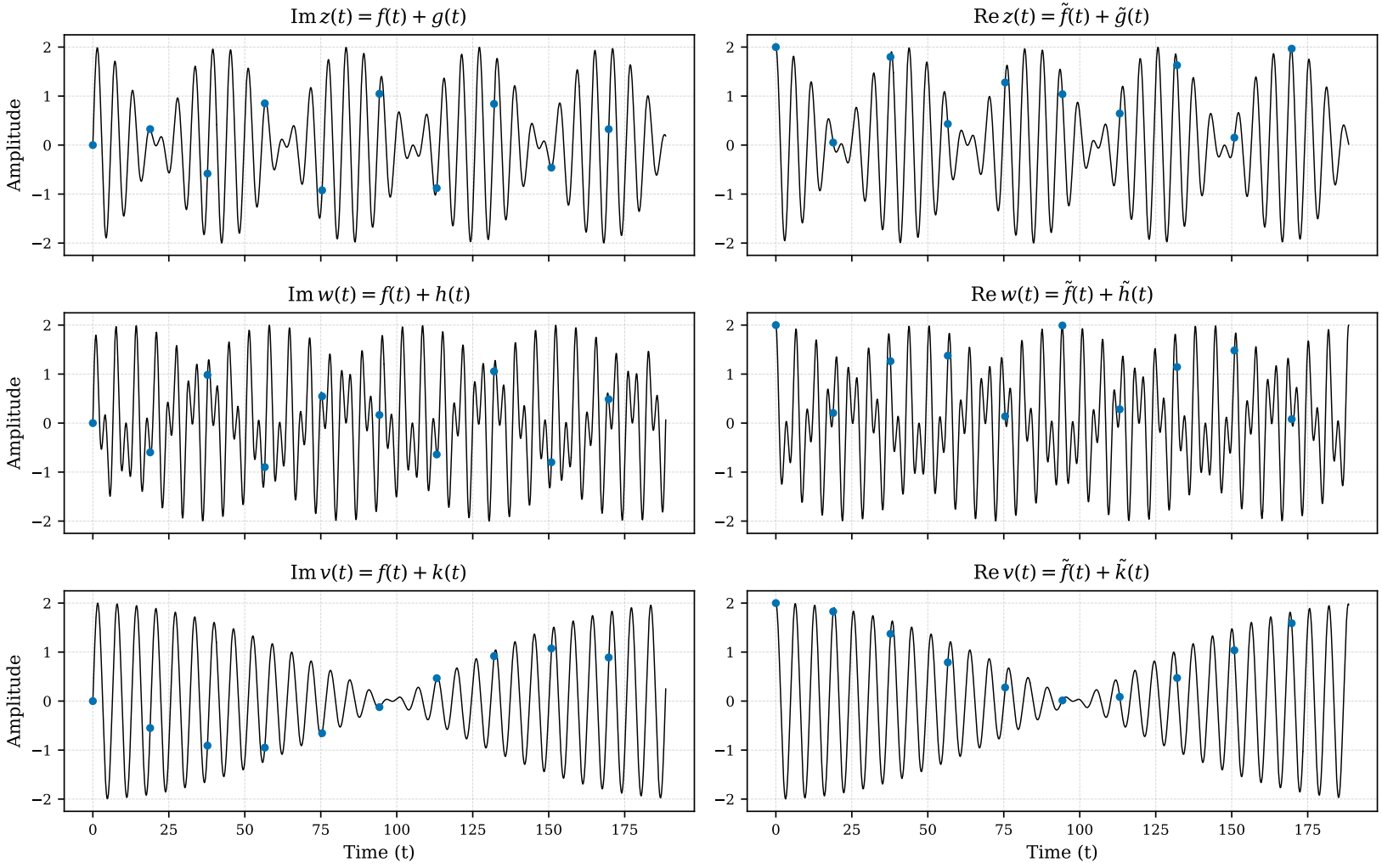


Figure 4: (cont.) (b) Imaginary and real parts of the resulting signals $z(t)$, $w(t)$, and $v(t)$, with their samples

This yields three real and three imaginary signals.

There are eight notational parameters: vertical notehead/stem position (pitch), microtonal accidentals, horizontal notehead/stem position (rhythm), amplitude of the dynamics curves (three interpolating curves per event), amplitude of the timbre gradients (three interpolating gradients per event), temporal position of the dynamics curves, temporal position of the timbre gradients, and left-hand touch pressure (for violin). See Section 2.2.3 and Appendix A for details. The pairings between the signals and the notational parameters are as follows:

Signal	**Notational Parameters**		
$\mathrm{Im}\, z(t)$	note/stem y-pos.	accidentals	
$\mathrm{Im}\, w(t)$	dynamics amp. 1	dynamics amp. 2	dynamics amp. 3
$\mathrm{Im}\, v(t)$	timbre grad. 1	timbre grad. 2	timbre grad. 3
$\mathrm{Re}\, z(t)$	note/stem x-pos.	L.H. touch press.	
$\mathrm{Re}\, w(t)$	dynamics pos. 1	dynamics pos. 2	dynamics pos. 3
$\mathrm{Re}\, v(t)$	timbre pos. 1	timbre pos. 2	timbre pos. 3

Table 1: The resultant signals and their notational parameters

Euler's formula serves not only for data augmentation but also to reflect an implication of the fourth melody: the fundamental unity between pitch, rhythm, and timbre. Timbre is a psychoacoustic phenomenon that can be formalized as a combination of periodic mechanical oscillations, each representing a pitch unfolding dynamically in time. The discrete perception of a periodic rhythmic event may transform into a continuous perception of pitch when it is repeated faster than roughly 20 times per second, the lower limit of pitch perception, demonstrating that rhythm *is* pitch.

The y-valued imaginary signal represents notational aspects that are conventionally associated with vertical motion, or with gradable adjectives such as "higher" and "lower": pitch, dynamics level, and harmonicity/inharmonicity. Similarly, the x-valued real signal represents those related to horizontal motion, namely rhythm and the temporal distribution of dynamics curves and timbre gradients. All these aspects are unified in one simple and elegant expression, $e^{i\theta}$. This expression is a part of the musical narrative as an unattainable ideal that constitutes the limit of the asymptotic relationship.

2.2.3 Mapping

To map the signals onto notational parameters, uniform sampling is performed on the signals (Figure 4). The sampled data is then organized into dictionaries:

```
real_z = sample_data[i]['re_z']
```

Mapping does not take place in a deterministic fashion—the value sampled from the signal is not readily assigned to a fixed notational value. The pitch parameter is defined as `y_data`:

```
y_data = np.random.normal(y_mean, y_std_dev, size)
```

Here, `np.random.normal` refers to the normal distribution sampling method of the NumPy library. The method takes three arguments: mean of the distribution (`y_mean`), standard deviation of the distribution (`y_std_dev`), and the number of samples (`size`). Since `y_data` corresponds to pitch, the mean of the distribution is the midpoint between the lowest and highest pitches of the instrument's range in the piece. This value stays fixed throughout the composition. Similarly, `size` is a predetermined value for each generation. The value sampled from the signal $\mathrm{Im}\, z(t)$ is used for `y_std_dev`, and affects the possible values to be drawn from the normal distribution.

The other parameters follow the same principle:

- **Rhythm.** The rhythm parameter is initially set up with evenly spaced positions, one per note. Each position is displaced by a value drawn from `np.random.normal(0, x_std_dev)`, where `x_std_dev` comes from $\mathrm{Re}\, z(t)$. The positions are then scaled and offset within the measure, kept in strictly increasing order, and separated by a minimum distance, so that a larger sample produces a more irregular rhythm without reordering the notes.

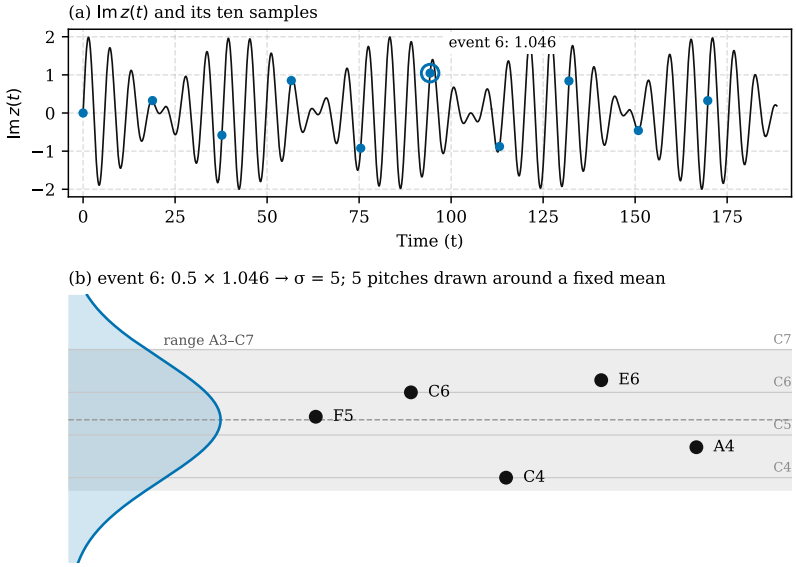


Figure 5: Statistical mapping of the signals onto the pitch parameter

- **Microtonal accidentals.** The angle of each pitch-alteration arrow is drawn from a normal distribution around the natural, with a spread derived from $\mathrm{Im}\, z(t)$. The same sample therefore governs how far the alterations spread, up to a semitone in either direction.
- **Left-hand touch pressure.** For the violin, touch values are drawn around a fixed mean with a spread derived from $\mathrm{Re}\, z(t)$.
- **Dynamics curves and timbre gradients.** All events except the shortest ones carry three Gaussian peaks for dynamics and three for timbre. The imaginary samples set their heights ($\mathrm{Im}\, w(t)$ for dynamics, $\mathrm{Im}\, v(t)$ for timbre), and the real samples set their positions within the event ($\mathrm{Re}\, w(t)$, $\mathrm{Re}\, v(t)$), each peak confined to one third of the event's length. The width of each peak is chosen from three ranges (narrow, middle, wide). The three peaks are summed into a single continuous curve, which is drawn as the dynamics curve or translated into the black-to-white timbre gradient. For short events, fewer than three peaks are used.

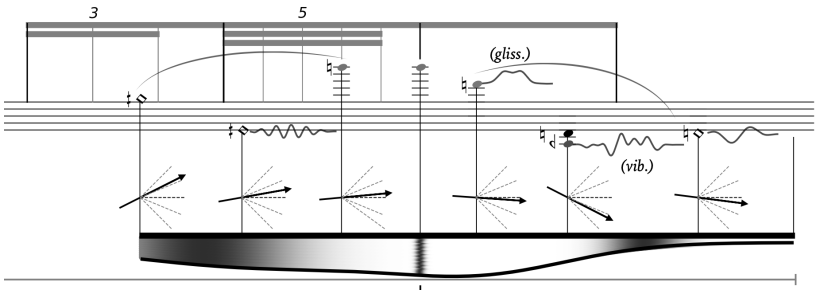


Figure 6: Dynamics curve and timbre gradients in the violin part

Each instrument has its own random seed, so a generation can be recalled and edited without changing its random draws.

2.3 Notation

La quatrième mélodie uses a notation that displays all musical parameters continuously, which at times contrasts starkly with the discrete nature of standard Western notation. Because commercially available notation software is impractical for this kind of notation, and because mappings such as those described above must be applied seamlessly, a custom Python (v. 3.12) codebase is used to generate and edit the notation. The notation is then exported as SVG files and edited further in Adobe Illustrator.

The performance notes of the piece provide detailed information on how to interpret the notational parameters. These notes can be found in Appendix A.

2.4 Intuitive layer

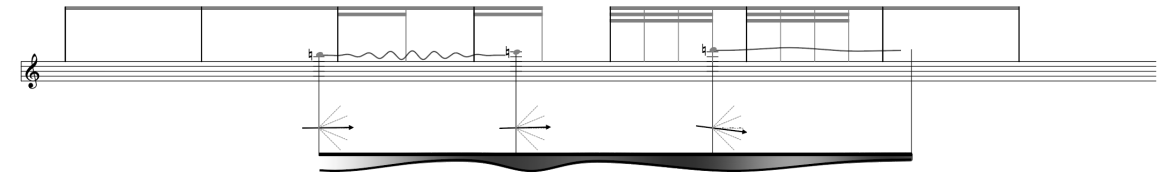
After a notation is generated, the music is shaped through an intuitive process. Its fleeting—and at times subconscious—nature makes this process difficult to document. The intuitive decisions are rather limited at the beginning of the piece, and gradually increase in number and complexity as the piece progresses. Additionally, they are not always grounded in a particular rationale, and at times driven solely by aesthetic preferences.

Table 2 below lists a few of the decisions that led from the program output in Figure 7(a) to the edited passage in Figure 7(b), which is heard in the first measure of the piece. This decision log does not stem from the original intuitive process; it is an attempt to retrace that process afterward, and as such it is incomplete.

Example #	**Decision**	**Reason**	**Action**
1	Make the tail of the *vibrato* a *portamento* move to the next pitch	It connects to the next pitch more gracefully this way	Edit the tail, add *port.*
2	Add *(vib.)* as well	*Vibrati* are written with larger curves than *portamenti*, so these two gestures need to be distinguished for clarity	Add *(vib.)*
3	No need for the extra flags	32nd resolution is not needed for the sustained tone	Remove the flags
4	D-natural should be E-quarter-flat	That simply sounds better to me	Edit the note and the microtonal angle accordingly

Table 2: Decision log for editing the generated notation in m. 1.

(a)



(b)

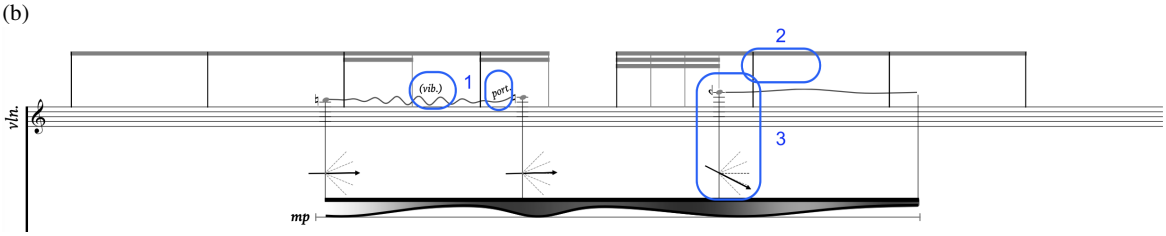


Figure 7: (a) Generated violin part; (b) the same passage after editing, with edits circled.

3 Performance

La quatrième mélodie was premiered by violinist Luke Fitzpatrick and percussionist Bonnie Whiting at the University of Washington on March 12, 2025. The performance is available online at https://www.yigitkolat.com/works/la_quatrieme_melodie/.

References

Durand, J.-F. (2006a). In the Mirror Land: Reflections on a Self-Reflection. In Bernard, J. W., editor, *Joël-François Durand in the Mirror Land*, pages 1–88. University of Washington Press, Seattle.

Durand, J.-F. (2006b). Melody—Three Situations: *Un Feu distinct*, *La Terre et le feu*, *Athanor*. In Bernard, J. W., editor, *Joël-François Durand in the Mirror Land*, pages 89–137. University of Washington Press, Seattle.

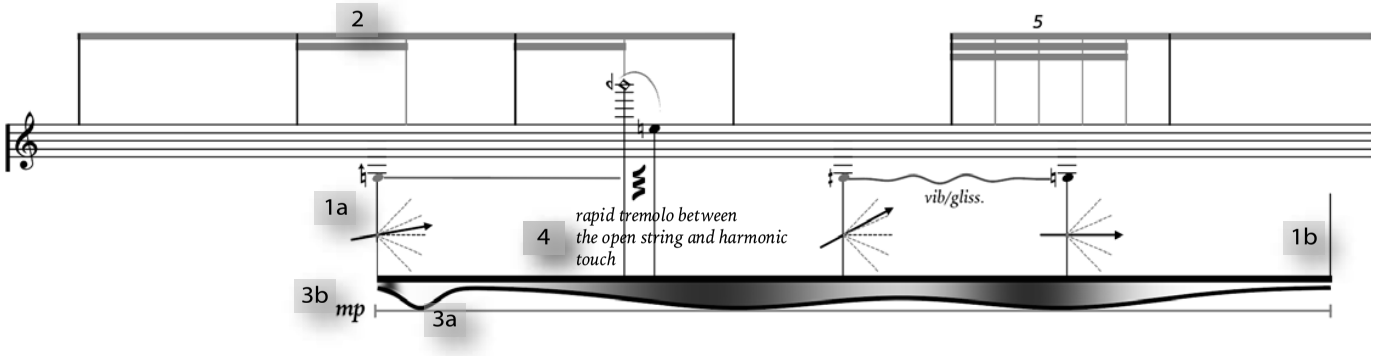
Durand, J.-F. (2020). *Geister, schwebende Geister...*: pour alto solo et ensemble. Score. Paris: BabelScores.

Durand, J.-F. (2026). Program note for *Quatuor à cordes no. 4, Musica fict(iv)a II*. In *Arditti Quartet* concert program, Meany Center for the Performing Arts, Seattle, pp. A-11–A-12. https://meanycenter.org/tickets/2026-10/production/arditti-quartet.

Appendix A Notation

Performance instructions:

Notes on notation for both performers



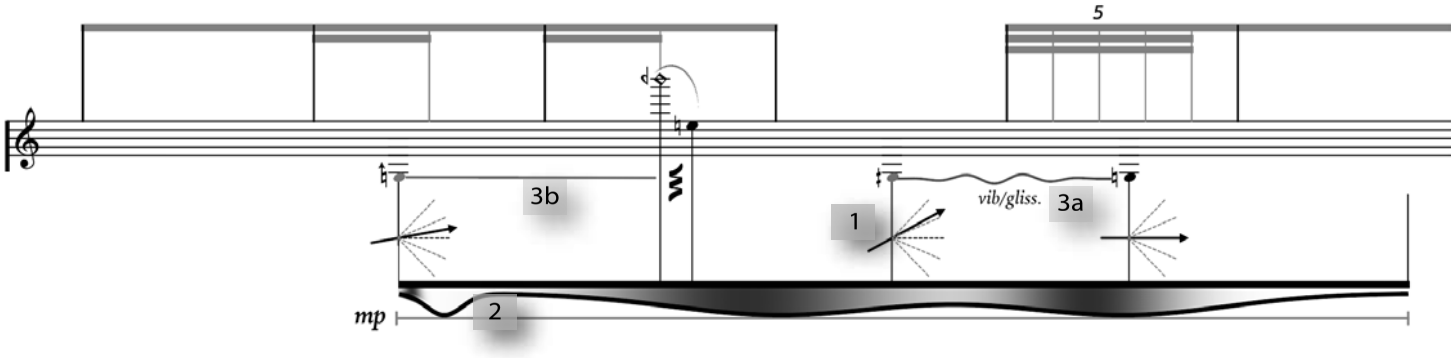
[1a] The notation features cells such as the one shown above. Each cell has a particular starting and ending point— everywhere else should be considered as "rest". All notes are sustained until the next note or the next vertical stem, e.g. the G3 at the end concludes by the headless stem **[1b]**.

[2] The grey rhythmic structures without noteheads show reference points and the actual rhythms written in black are determined based on their proximity to them. The eighth beats are articulated with black stems.

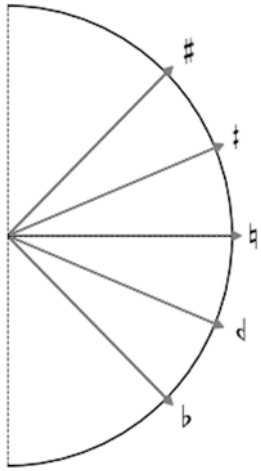
[3a] The curve that extends downward is the dynamics curve: the farther it opens, the more physical power is applied by the performer. Note that this doesn't necessarily always equate to a louder result — it simply means a considerable power is applied by the performer, which might not yield an analogous acoustic response. The conventional dynamics indications with a line tangential to the curve serves as a reference for the dynamics level **[3b]**.

[4] Some of the techniques/symbols are explained directly on the score.

Notes on notation for the violin part



[1] This shape represents pitch alterations. The straight black arrow symbolizes "natural", and the rest of the conventional accidentals can be thought graphically as angles formed upon the natural, as seen in the graph on the right.

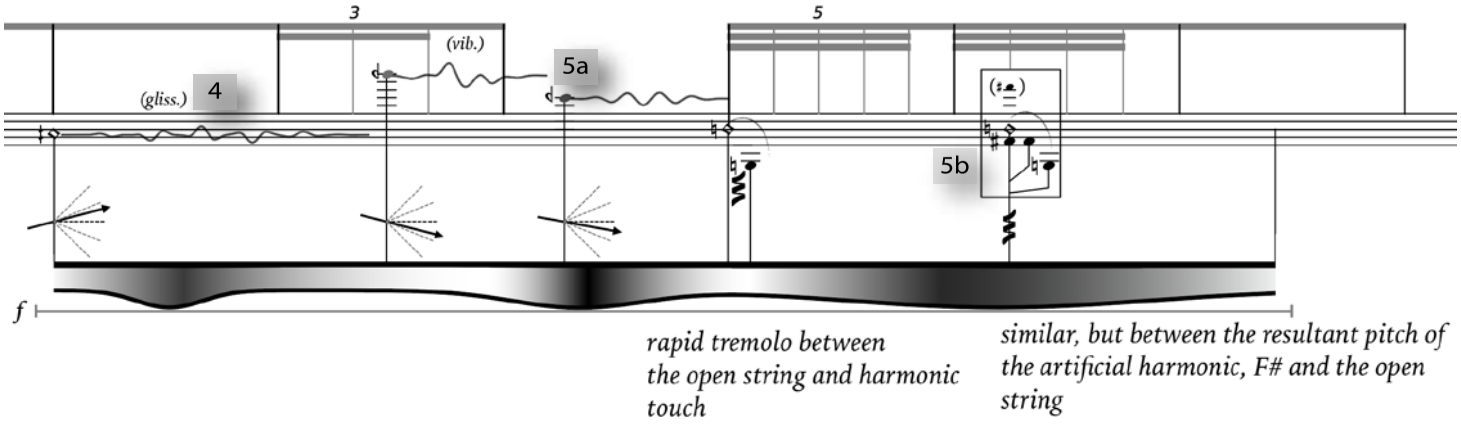


Often the arrow falls somewhere between these angles **[1]**. For these, an approximate accidental is given, but the fine-tuning of the pitch is completely left to the performer's discretion: for the situations where it appears to be too impractical, the written accidental can be played.

[2] The black-white gradient above the dynamics curve is the timbre gradient. The black hue denotes higher bow pressure with *molto sul ponticello*, whereas white hue stands for *flautando* with *ordinario* vertical bow position (not *sul tasto*). The bow pressure should never completely overpower the pitch content in black areas.

[3a] Vibrato always appears as a simple sinusoidal wave or as some variation of it. The amplitude and frequencies are drawn in relation to each other, i.e. one vibrato can be faster/deeper than the other than another, but no absolute vibrato frequency or amplitude value is expected. Please note that they are drawn rather large for clarity.
When there is no line on a pitch that is not one of the open strings, the "default" form of vibrato can be used. *Non vibrato* is notated with a straight line **[3b]**.

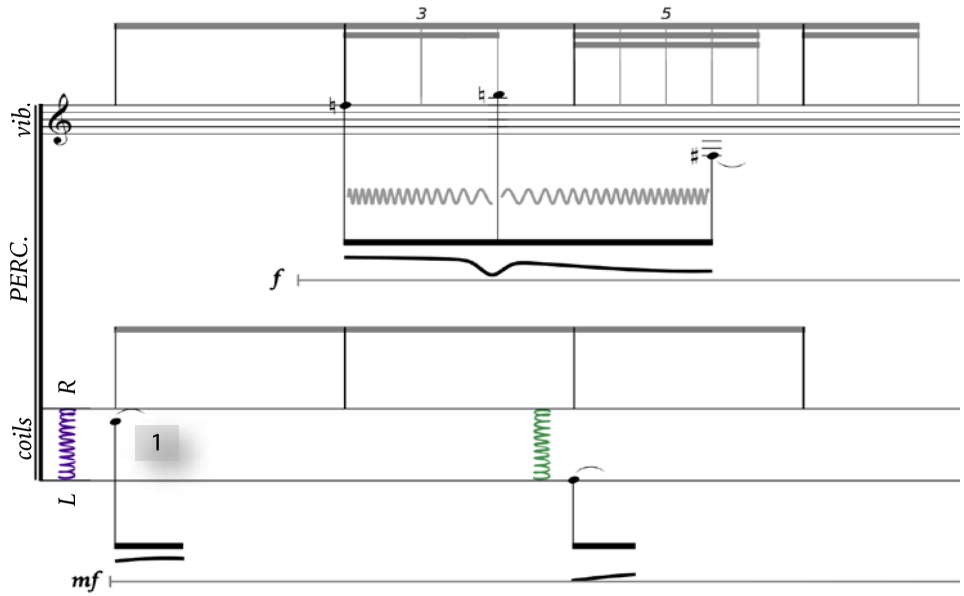
Notes on notation for the violin part (cont'd)



[4] Non-sinusoidal shapes are pitch "meanderings", their range of motion are larger than the vibrati, even though they are drawn at the same scale. In most cases they are distinguished from the vibrati with a *(gliss.)* note next to them. These can be interpreted rather freely.

[5a] There are black-to-light grey gradients for the L. H. touch pressure as well. The interpretation of these can range from just three shades --*black = normal pressure, grey = half-harmonic pressure, diamond shape = harmonic pressure*-- to a any finer "resolution". Harmonic touches are distinguished by diamond shapes. The resultant pitches on the harmonic nodes are not notated and their clarity is not particularly expected except one passage seen in the example here **[5b]**.

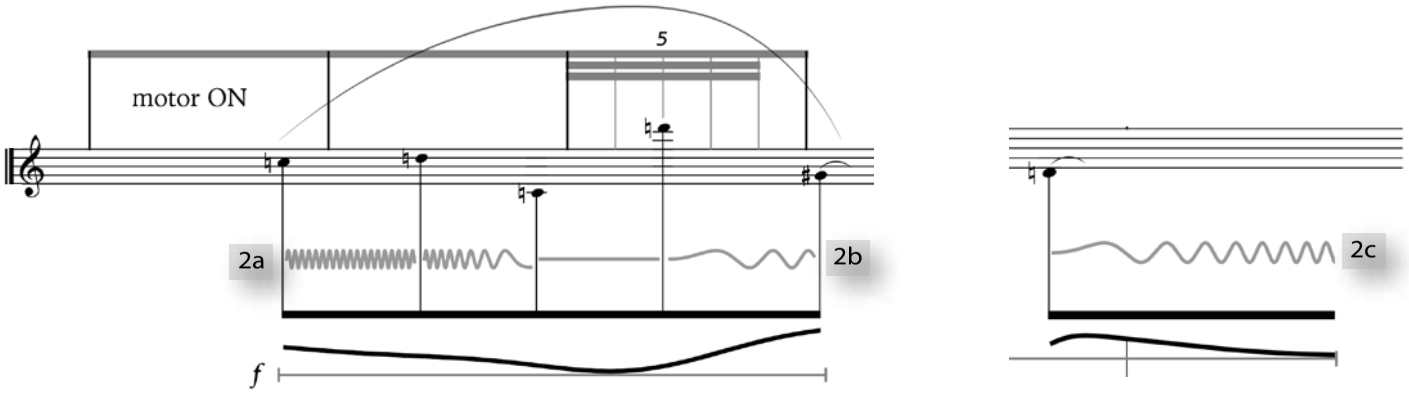
Notes on the percussion part



The percussion part is written for a vibraphone and 2 steel spring coils. The spring coils are placed on a resonant surface such as the surface of a bass drum. The protective layer between the coils and the drums should have minimal impact on reverberation. For the coils, metal mallets and hard yarn mallets are recommended.

The vibraphone and the coils appear on different staves. A clef that resembles the shape of a spring coil is used to denote the attack points on the instrument [**1**]. The coils clef appears in two colors, which can be arbitrarily assigned to the instruments. However, once the colors are assigned to their respective instruments, the assignment should not change during the performance.

Notes on the percussion part (cont'd)



[2a] Vibrato is a crucial component in this piece, and the notation asks for frequent changes in vibrato. These changes are notated graphically. In cases where R.H. is used both for striking the bar and changing the vibrato, the change can be made whenever R.H. is back to the position --no continuous change is expected.

All percussion events end with *l.v.*, because of this, the last vibrato value before the final stem **[2b]** stays unchanged as the sound fades. In some cases, the vibrato is changed while the sound fades away **[2c]**. It is explicitly indicated in the score when vibrato remains unchanged.